

COMPLEXITY AND RECURRENCES

Algorithmic Performance and Mathematical Analysis

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Course Outline

1 Introduction to Complexity Analysis

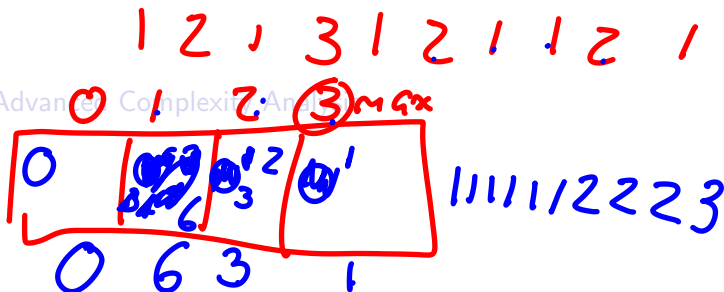
2 Advanced Complexity Analysis



Outline

1 Introduction to Complexity Analysis

2 Advanced Complexity Analysis



Why Analyze Algorithms?

The Fundamental Question

Given multiple algorithms that solve the same problem, which one should we choose?

Counting Sort $\rightarrow$ repetitions

① Max $\rightarrow$ number of list

②  $\rightarrow$ $len = max + 1$

③ for ∇ in numbers:
 $result[len] += 1$

④ print $result[i] \rightarrow i$

Why Analyze Algorithms?

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Given multiple algorithms that solve the same problem, **which one should we choose?**

- **Performance:** How fast does the algorithm run?
- **Resources:** How much memory does it use?
- **Scalability:** How does it handle large inputs?
- **Predictability:** Can we predict its behavior?

contest

Real World Impact

Condor

- User experience depends on algorithm speed
- Server costs increase with inefficient algorithms
- Mobile apps drain battery with poor algorithms
- Scientific computing requires efficient solutions



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- Scientific computing requires efficient solutions

Elementary Operations Identification I

Definition

An **elementary operation** is a basic computation that takes constant time regardless of input size.

Common Elementary Operations:

• Assignment: `x = 5`

• Arithmetic: `+`, `-`, `*`, `/`

`<`, `>`, `<=`, `>=`

• Access: `array[1]`

• Call: `func()` Simple built-in functions

time → processor



Elementary Operations Identification I

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An **elementary operation** is a basic computation that takes **constant time** regardless of input size.

Common Elementary Operations:

① step

- **Assignment:** $x = 5$
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- **Comparison:** $==$, $<$, $>$
- **Array Access:** $\text{array}[i]$
- **Function Calls:** Simple built-in functions



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① each one

$x + y - z \Rightarrow 2 \text{ steps}$



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- **Function Calls:** Simple built-in functions] (v) call
↳ sum inner steps



Elementary Operations Identification II

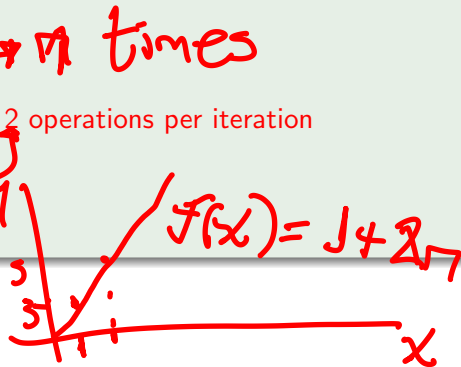
Example

Count operations in simple code:

```

1:  $sum = 0$  // 1 operation
2: for  $i = 0$  to  $n - 1$  do
3:    $sum = sum + array[i]$  // 2 operations per iteration
4: end for
  
```

Total: $1 + 2n$ operations



Study Case: Counting Operations in Arrays Union I

Problem

Given two arrays A and B, create array C that contains all elements from both arrays without duplicates.

1: Copy all elements from A to C

2: **for** each element in B

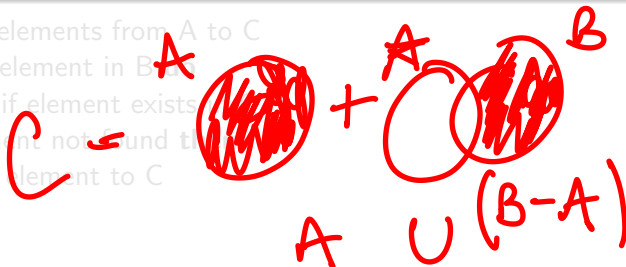
3: Check if element exists

4: **if** element not found then

5: Add element to C

6: **end if**

7: **end for**



Study Case: Counting Operations in Arrays Union I

Problem

Given two arrays A and B, create array C that contains all elements from both arrays without duplicates.

- 1: Copy all elements from A to C
- 2: **for** each element in B **do**
- 3: Check if element exists in C
- 4: **if** element not found **then**
- 5: Add element to C
- 6: **end if**
- 7: **end for**

$C[i] = A[i]$
 $\text{len}(A) \rightarrow 2$
 $\text{len}(B) = 1$

$\boxed{3}$



Study Case: Counting Operations in Arrays Union II

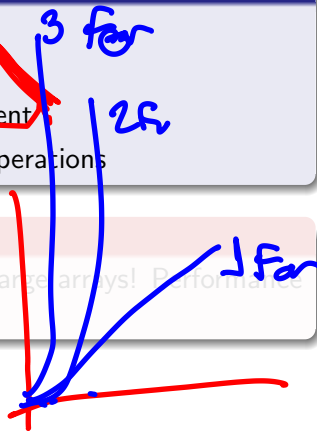
Operation Analysis

- Copy from A: n operations
- For each element in B: m iterations
- Search in C: Up to n comparisons per element
- Total worst case: $n + m \cdot n = n(1 + m)$ operations

Key Insight

The nested search makes this algorithm slow for large arrays! Performance grows quadratically.

↓ for ∇^2
 2 for ∇^2
 3 for ∇^3



Study Case: Counting Operations in Arrays Union II

Operation Analysis

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- **Search in C:** Up to n comparisons per element
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Best, Average, Worst Case Scenarios

Three Perspectives on Algorithm Performance

Different inputs can lead to dramatically different execution times for the same algorithm.

Best Case

- Minimum operations
- Most favorable input
- Example: Finding element at first position

Average Case

- Expected performance
- Typical input
- Example: Finding element in middle

Worst Case

- Maximum operations
- Least favorable input
- Example: Element not found



Best, Average, Worst Case Scenarios

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- Minimum operations
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→ sort → sorted array

binary → element middle

/lined

Average Case

- Typical input
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- Example: Finding element in middle

binary → random
 sort → 50%
 random
 linear



Best, Average, Worst Case Scenarios

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- Example: Element not found

sort → sorted reverse

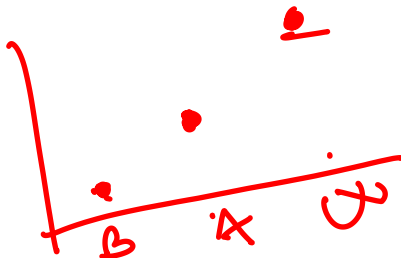


Best, Average, Worst Case Scenarios II

Example

Search in array [5, 2, 8, 1, 9]:

- **Best:** Search for 5 $\rightarrow$ 1 comparison
- **Average:** Random element $\rightarrow \sim 2.5$ comparisons
- **Worst:** Search for 7 (not found) $\rightarrow 5$ comparisons



Performance Prediction Methods

Approaches to Predict Algorithm Performance

- 1 **Theoretical Analysis:** Mathematical analysis
- 2 Empirical Analysis: Timing actual executions
- 3 Simulation: Testing with synthetic data
- 4 Profiling: Detailed runtime analysis



Performance Prediction Methods

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correlation



Performance Prediction Methods

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- 1 **Theoretical Analysis:** Mathematical analysis
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time



Performance Prediction Methods

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Performance Prediction Methods II

Theoretical Analysis

Advantages:

- Machine-independent
- Covers all possible inputs
- Mathematical precision

Limitations:

- Ignores implementation details
- May not reflect real hardware

Empirical Analysis

Advantages:

- Reflects real performance
- Shows actual constants
- Platform-specific results

Limitations:

- Machine-dependent
- Limited input cases
- Time-consuming

Best Practice

Use theoretical analysis for algorithm design, then validate with empirical testing.

GPU/Quadcore

i.e. $\rightarrow$ 200 minutes

steps

$f(n) \rightarrow n = 10$

100
1000

$t \approx 100.000.000$

Performance Prediction Methods II

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Time vs. Space Trade-offs

security vs speed

Definition

A time-space trade-off occurs when we can reduce time by using more memory, or reduce memory at the cost of increased time.

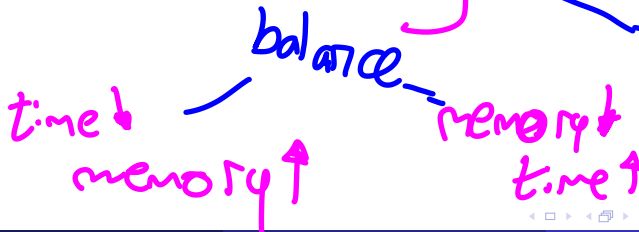
Trade-off Decision Factors:

- Available memory constraints
- Time requirements
- Cost of computation vs. storage
- Frequency of algorithm execution

time $\sim$ processor

space $\sim$ memory

constraint



Time vs. Space Trade-offs I

Definition

A **time-space trade-off** occurs when we can **reduce time** by using more memory, or **reduce memory** at the cost of increased time.

Trade-off Decision Factors:

- Available memory constraints ?
- Time requirements ?
- Cost of computation vs. storage /
- Frequency of algorithm execution

Handwritten notes illustrating trade-offs:

500 ms
 ~ 1 ms } 4s

real-world



Time vs. Space Trade-offs II

Example

Fibonacci Calculation:

Space-Efficient (Recursive):

- Uses minimal extra memory
- But recalculates same values repeatedly
- Very slow for large numbers

Time-Efficient (Memoization): →

- Stores previously calculated values
- Uses more memory
- Much faster for large numbers

vectorization



Systematic Complexity Calculation Methods I

Step-by-Step Approach

- ① **Identify:** All loops and recursive calls
- ② **Count:** Operations in each section
- ③ **Multiply:** Operations by execution frequency
- ④ **Sum:** All contributions
- ⑤ **Simplify:** Focus on dominant terms

quadratic
linear

$2n$
 $3n$
 n^2

$$3n^3 + 2 \Rightarrow 3n^3 \rightarrow n^3$$

$$x^2 + x + 1 \Rightarrow x^2 \rightarrow \text{dominant}$$



Systematic Complexity Calculation Methods II

Example

Nested Loop Analysis:

```

1: for  $i = 0$  to  $n - 1$  do
2:   for  $j = 0$  to  $n - 1$  do
3:     result = array[ $i$ ] + array[ $j$ ]
4:   end for
5: end for
  
```

$$n * n * 4$$

$$4n^2$$

$$\times$$

Operation Count:

- Outer loop runs n times
- Inner loop runs n times for each outer iteration
- Total operations: $n * n = n^2$ operations

word



Study Case: Complexity Calculation for QuickSort I

QuickSort Algorithm Structure

- 1: Choose a **pivot** element
- 2: **Partition** array around pivot
- 3: Recursively sort **left** part
- 4: Recursively sort **right** part



Partition Operation

- Compare each element with pivot
- Rearrange elements (smaller left, larger right)
- This takes n operations for n elements

No quadratic



Study Case: Complexity Calculation for QuickSort I

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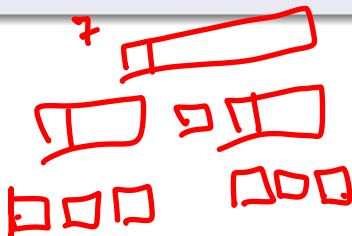
- **Compare** each element with pivot $7 \rightarrow 1 \rightarrow 7$
- **Rearrange** elements (smaller left, larger right) $1 \rightarrow 7 \rightarrow 7$
- This takes **n operations** for n elements



Study Case: Complexity Calculation for QuickSort II

Best Case

- Pivot divides array in half
- Balanced partitions ✓
- Fast performance



Worst Case

- Pivot is always smallest/largest
- Unbalanced partitions
- Slow performance



Outline

1 Introduction to Complexity Analysis

2 Advanced Complexity Analysis



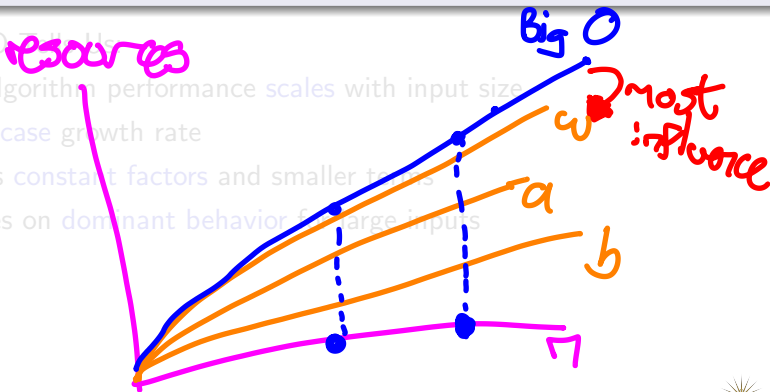
Big O Notation: Definition and Meaning

Definition

Big O notation describes the upper bound of an algorithm's growth rate for large inputs.

What Big O Tells Us

- How algorithm performance scales with input size
- Worst-case growth rate
- Ignores constant factors and smaller terms
- Focuses on dominant behavior for large inputs



Big O Notation: Definition and Meaning

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What Big O Tells Us:

- How algorithm performance **scales** with **input size**
- **Worst-case** growth rate
- Ignores **constant factors** and smaller terms
- Focuses on **dominant behavior** for large inputs

$$x^7 + c_x$$

$$x^7$$

dominant factor

relate with
↑ Big O



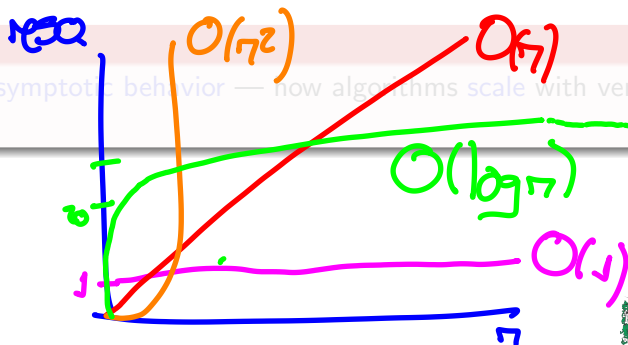
Big O Notation: Classifications

Common Big O Classifications:

- $O(1)$ — **Constant**: Always same time
- $O(n)$ — **Linear**: Time grows with input size
- $O(n^2)$ — **Quadratic**: Time grows with square of input
- $O(\log n)$ — **Logarithmic**: Very efficient growth

Key Insight

Big O focuses on asymptotic behavior — how algorithms scale with very large inputs.



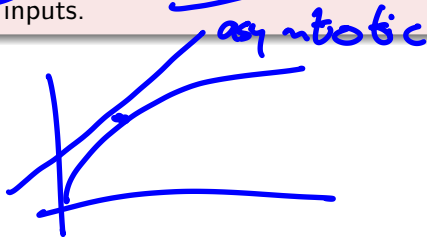
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Upper Bound Concept and Mathematical Definition

$$a^b = c \rightarrow \log_a c = b$$

$$\log 7 \Rightarrow \log_2 7$$

Understanding Upper Bounds

An upper bound guarantees that an algorithm will (never perform worse) than the bound.

Big O Meaning:

- Sets a ceiling on performance
- Guarantees maximum time
- Used for worst-case analysis
- Helps compare algorithms

Common Growth Orders:

- $O(1)$ — Constant
- $O(\log n)$ — Logarithmic
- $O(n)$ — Linear
- $O(n \log n)$ — Linearithmic
- $O(n^2)$ — Quadratic
- $O(2^n)$ — Exponential



common O&C



Study Case: First Letter in a Sentence Complexity

Problem

Find the **first alphabetical letter** in a given sentence string.

```

1: for each character in sentence do
2:   if character is a letter then
3:     return character ✓
4:   end if
5: end for
6: return null

```

"zzzzz"
cont.
alphabet

Best: First character is a letter

- Only check 1 character
- Constant time: $O(1)$

all characters are not
in the alphabet

Worst: No letters in sentence

- Check every character
- Linear time: $O(n)$

Even though best case is $O(1)$, we report $O(n)$ because Big O represents the **worst-case upper bound**.

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```

Best: First character is a letter

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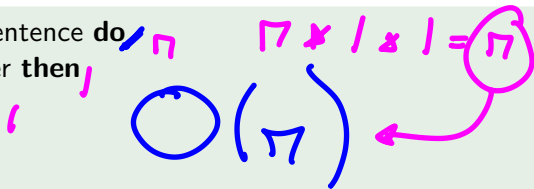
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- Check every character
- Linear time: $O(n)$

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Find the **first alphabetical letter** in a given sentence string.

```
6: return null
```



Worst: No letters in sentence

- Linear time: $O(n)$

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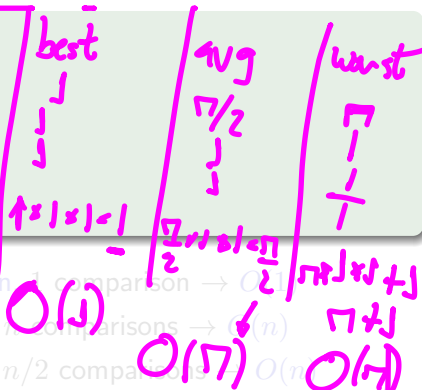
Study Case: Linear Search Algorithm Complexity

Problem

Search for a **target element** in an **unsorted** array.

```

1: for each element in array do
2:   if element equals target then
3:     return index
4:   end if
5: end for
6: return -1
  
```



- **Best Case:** Element at first position, 1 comparison $\rightarrow O(1)$
- **Worst Case:** Element not in array, n comparisons $\rightarrow O(n)$
- **Average Case:** Element in middle, $n/2$ comparisons $\rightarrow O(n)$



Study Case: Linear Search Algorithm Complexity

Problem

Search for a **target element** in an **unsorted** array.

```
1: for each element in array do  
2:   if element equals target then  
3:     return index  
4:   end if  
5: end for  
6: return -1
```

- **Best Case:** Element at **first position**, 1 comparison $\rightarrow O(1)$
- **Worst Case:** Element **not in array**, n comparisons $\rightarrow O(n)$
- **Average Case:** Element in **middle**, $n/2$ comparisons $\rightarrow O(n)$



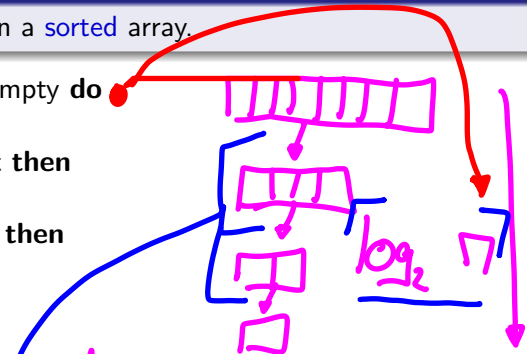
$$L_1 = O(\log n)$$

Search for a target element in a **sorted** array.

- ```

1: while search space not empty do
2: Find middle element
3: if middle equals target then
4: return index
5: else if middle > target then
6: search right half
7: else
8: search left half
9: end if
10: end while
11: return -1

```



$a^b = c \rightarrow \log_a c = b$

$2^3 = 8 \rightarrow \log_2 8 = 3$

$$\log_2 7 \Rightarrow \log 7$$



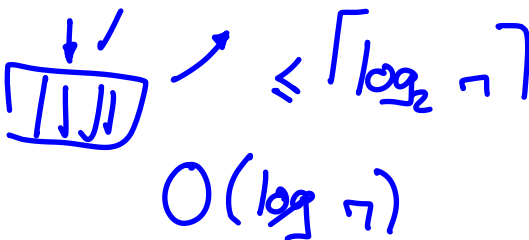
# Study Case: Binary Search Algorithm Complexity II

Each step eliminates half of the remaining elements:

- Step 1:  $n$  elements
- Step 2:  $n/2$  elements
- Step 3:  $n/4$  elements
- Step  $k$ :  $n/2^k$  elements

base 2

Algorithm stops when we reach 1 element.



# Big Omega ( $\Omega$ ) and Big Theta ( $\Theta$ ) Notations I

## Big Omega ( $\Omega$ )

### Lower Bound:

- Minimum time required
- Best-case analysis
- “At least this fast”
- Guarantees algorithm won't be faster

## Big Theta ( $\Theta$ )

### **Tight Bound:**

- Exact growth rate
- Best and worst are same
- “Exactly this fast”
- Most precise description

re2



# Big Omega ( $\Omega$ ) and Big Theta ( $\Theta$ ) Notations II

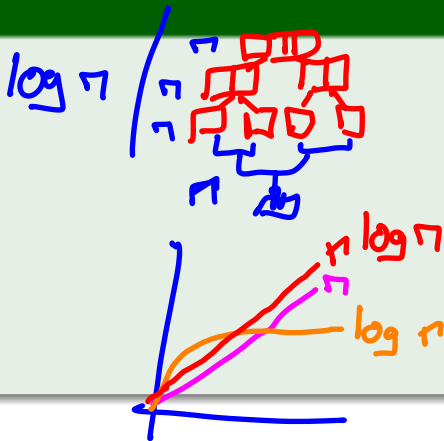
## Example

### Merge Sort:

- Best case:  $\Omega(n \log n)$  /
- Worst case:  $O(n \log n)$  /
- Since best = worst:  $\Theta(n \log n)$

### Linear Search:

- Best case:  $\Omega(1)$
- Worst case:  $O(n)$
- No tight bound (best  $\neq$  worst)



# Study Case: Tower of Hanoi Problem I

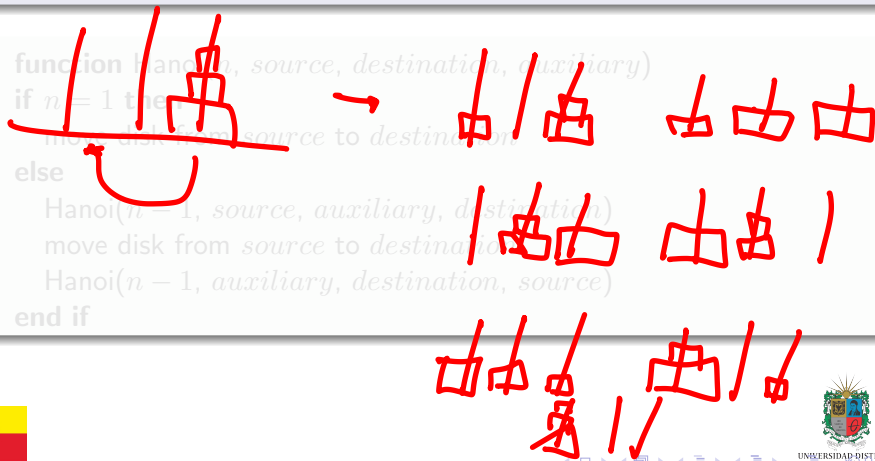
## Problem

Move  $n$  disks from **source** rod to **destination** rod using **auxiliary** rod. Rules: only move **one disk** at a time, never place **larger disk** on smaller one.

```

1: function Hanoi(n, source, destination, auxiliary)
2: if n = 1 then
3: move disk from source to destination
4: else
5: Hanoi(n - 1, source, auxiliary, destination)
6: move disk from source to destination
7: Hanoi(n - 1, auxiliary, destination, source)
8: end if

```



# Study Case: Tower of Hanoi Problem I

## Problem

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```

1: function Hanoi(n , source, destination, auxiliary)
2: if $n = 1$ then
3: move disk from source to destination
4: else
5: Hanoi($n - 1$, source, auxiliary, destination) 1 → 7
6: move disk from source to destination 1
7: Hanoi($n - 1$, auxiliary, destination, source) 1 → 7
8: end if

```

*base*  
*recursive*



- $T(1) = 1$  (base case: move one disk)
- $T(n) = 2T(n-1) + 1$  (two recursive calls + one move)



- $T(n) = 2^n - 1$
- Complexity:  $O(2^n)$  — exponential!



# Study Case: Tower of Hanoi Problem II

## Recurrence Analysis

- $T(1) = 1$  (base case: move one disk)
- $T(n) = 2T(n - 1) + 1$  (two recursive calls + one move)

## Solution:

- $T(n) = 2^n - 1$
- Complexity:  $O(2^n)$  — exponential!

## Real-World Implication

For 64 disks:  $2^{64} - 1 \approx 18 \text{ billion}$  moves!



# Solving Recurrence Relations

## Common Solution Methods

- ① **Substitution Method:** Guess and verify
- ② **Recursion Tree:** Visualize recursive calls
- ③ **Master Theorem:** Direct formula for common cases
- ④ **Characteristic Equation:** For homogeneous recurrences

*Empirical*  
*D4Q*  
*Prints*

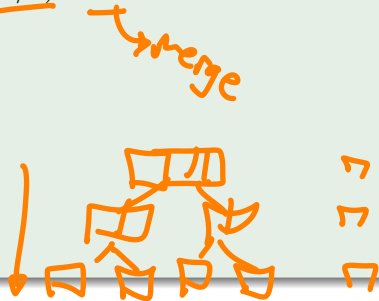


# Solving Recurrence Relations

## Example

**Recursion Tree Method for  $T(n) = 2T(n/2) + n$ :**

- Level 0:  $n$  work
- Level 1:  $2 \times (n/2) = n$  work
- Level 2:  $4 \times (n/4) = n$  work
- ...
- Total levels:  $\log n$
- Total work:  $n \times \log n = O(n \log n)$



## Method Selection

- Simple cases: Direct solving
- Divide-and-conquer: Master theorem
- Complex cases: Substitution method

# Solving Recurrence Relations

## Example

### Recursion Tree Method for $T(n) = 2T(n/2) + n$ :

- Level 0:  $n$  work
- Level 1:  $2 \times (n/2) = n$  work
- Level 2:  $4 \times (n/4) = n$  work
- ...
- Total levels:  $\log n$
- Total work:  $n \times \log n = O(n \log n)$

## Method Selection

- Simple cases: Direct solving ✓
- Divide-and-conquer: Master theorem ✓
- Complex cases: Substitution method ] guess

# Master Theorem Introduction I

## Master Theorem

For recurrences of the form  $T(n) = \boxed{T(n/b)} + \boxed{f(n)}$ :

**Case 1:**  $f(n)$  grows **slower** than  $n^{(\log_b a)}$

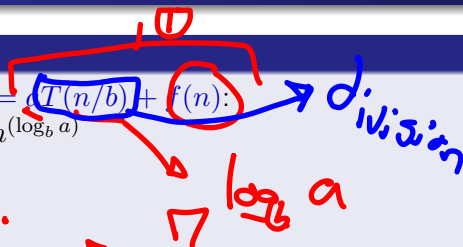
- $T(n) = O(n^{(\log_b a)})$  ✓

**Case 2:**  $f(n)$  grows at **same rate**.

- $T(n) = O(n^{(\log_b a)} \log n)$  —

**Case 3:**  $f(n)$  grows **faster**

- $T(n) = O(f(n))$  ✓



# Master Theorem Introduction II

## Examples

### Merge Sort:

- $T(n) = 2T(n/2) + n$
- $a = 2, b = 2, f(n) = n$
- Case 2:  $O(n \log n)$

### Binary Search:

- $T(n) = T(n/2) + 1$
- $a = 1, b = 2, f(n) = 1$
- Case 2:  $O(\log n)$

When to Use

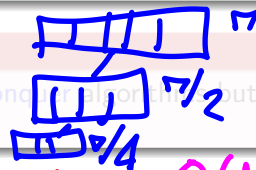
Master theorem works for many divide-and-conquer algorithms but has limitations.



$$n^{\log_2 2}$$



$$n \rightarrow n \rightarrow O(n \cdot \log n)$$



$$n^{\log_2 1}$$

$$n \rightarrow 1 = f(n)$$

$$O(1 \cdot \log n) \rightarrow O(\log n)$$



# Master Theorem Introduction II

## Examples

### Merge Sort:

- $T(n) = 2T(n/2) + n$
- $a = 2, b = 2, f(n) = n$
- Case 2:  $O(n \log n)$

### Binary Search:

- $T(n) = T(n/2) + 1$
- $a = 1, b = 2, f(n) = 1$
- Case 2:  $O(\log n)$

## When to Use

Master theorem works for many divide-and-conquer algorithms but has limitations.



# Recursive Algorithm Complexity Calculation I

## Systematic Approach

① **Identify:** Base case(s) and recursive case(s)

② **Set up:** Recurrence relation

$\rightarrow O(J)$  or  $2(T/2)$  or  $2(T-1)$

③ **Solve:** Using appropriate method

④ **Verify:** Check solution makes sense

for example

experiments  
↓  
exploration  
size



# Recursive Algorithm Complexity Calculation II

## Merge Sort Analysis

### Step 1: Identify cases

- Base: Array size 1  $\rightarrow O(1)$  *ignore*
- Recursive: Divide, sort halves, merge

### Step 2: Set up recurrence

- $T(n) = 2T(n/2) + O(n)$

### Step 3: Solve using Master Theorem

- Case 2:  $T(n) = O(n \log n)$

### Step 4: Verify

- Makes sense: logarithmic levels, linear work per level



# Summary I

## Key Takeaways

- **Complexity Analysis:** Essential for algorithm comparison and selection
- **Big O Notation:** Describes growth rate for large inputs
- **Recursion:** Powerful technique requiring careful base case design
- **Recurrence Relations:** Mathematical tools for recursive algorithm analysis

- substitution
- recursion tree



# Summary II

## Practical Skills Gained

- Count operations systematically ✓
- Classify algorithms by growth rate ✓
- Design recursive solutions correctly ✓
- Analyze recursive algorithm complexity ✓
- Apply Master Theorem to common patterns ✓

## Next Steps

- Practice complexity analysis on various algorithms
- Implement recursive solutions with proper optimization
- Study advanced recurrence solving techniques
- Apply knowledge to real-world algorithm selection



# Summary II

## Practical Skills Gained

- Count operations systematically
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## Next Steps

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# Outline

- 1 Introduction to Complexity Analysis
- 2 Advanced Complexity Analysis



# Thanks!

## Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-i>

